

Why Product Metrics Improve While the Product Feels Worse

Proxy Optimization, Experience Drift, and User Experience

Reality Drift Framework — Short Concept Note #5

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The Basic Pattern

Product teams cannot observe every user experience directly.

They rely on representations.

Conversion rates represent success.

Retention represents value.

Engagement represents interest.

Onboarding completion represents understanding.

Customer satisfaction scores represent sentiment.

At first, these indicators help teams identify real problems and improve the product.

Over time, the indicators become easier to optimize than the experience they were created to represent.

The core sequence is:

user experience → measurement → target → optimization → proxy substitution → experience drift

A Metric Is a Representation of Experience

A product metric does not measure the entire experience.

It captures one observable feature of it.

A completed onboarding flow may indicate that the user reached the final screen.

It does not necessarily show that the user understands the product.

A high retention rate may indicate that people continue returning.

It does not necessarily show that they value the experience.

A rising engagement rate may reflect interest.

It may also reflect confusion, habit, dependency, or deliberate friction.

Metrics make experience legible.

They do not preserve all of its meaning.

Why Product Teams Need Proxies

Direct experience is difficult to manage at scale.

A team cannot watch every customer use the product.

It cannot interview every person after every interaction.

It cannot directly observe trust, frustration, comprehension, or long-term value across millions of users.

Metrics solve this problem.

They compress complex experiences into signals that can be compared across time.

The signals make prioritization possible.

They help teams recognize patterns.

They allow organizations to coordinate around shared evidence.

The problem does not begin with measurement.

It begins when the representation becomes the operational definition of success.

Proxy Optimization

Proxy Optimization occurs when a system improves the measurable indicator rather than the underlying condition the indicator was designed to represent.

The metric begins as evidence.

It becomes a target.

The team learns which interventions move it.

Product decisions increasingly optimize the observable number.

The number improves.

The experience may not.

This does not always involve manipulation.

Teams can sincerely improve a metric while weakening the broader outcome.

The proxy responds faster than the reality behind it.

When the Metric Becomes the Product

A product team may begin with a user problem.

Over time, the work becomes organized around measurable outcomes.

Improve activation.

Increase session length.

Reduce abandonment.

Raise conversion.

Increase feature adoption.

These goals sound user-centered because they were originally derived from user behavior.

But once operationalized, the organization can begin treating metric movement as proof of improvement.

The product is no longer judged primarily by the experience it creates.

It is judged by the indicators it produces.

Experience Drift

Experience Drift occurs when representations of user experience continue improving while lived experience gradually moves elsewhere.

The dashboards remain healthy.

The reports remain positive.

Targets are reached.

Experiments produce statistically significant gains.

Yet users describe the product as more confusing, intrusive, exhausting, or difficult to trust.

The system remains operationally successful.

The experience becomes progressively less visible through the measurements used to manage it.

Why Optimization Narrows the Experience

Every metric selects one feature of reality.

Optimization increases attention to that feature.

What is measurable becomes visible.

What is visible becomes actionable.

What is actionable receives resources.

Other parts of the experience become secondary.

A team optimizing onboarding completion may reduce opportunities for exploration.

A team optimizing engagement may increase notifications and interruptions.

A team optimizing conversion may make refusal more difficult.

A team optimizing support resolution time may discourage complex cases.

The chosen metric does not merely describe the product.

It directs the product's evolution.

Local Improvement and Global Decline

Product teams often optimize individual touchpoints.

The sign-up flow becomes faster.

The checkout process becomes shorter.

The support interaction becomes more standardized.

The notification system becomes more effective.

Each component improves according to its local metric.

The overall experience can still become worse.

A smoother sign-up process may create expectations the product cannot fulfill.

A more efficient support workflow may make unusual problems impossible to explain.

A higher-performing notification system may increase interruption across the user's day.

The parts improve.

The journey fragments.

Why Users and Dashboards Disagree

Dashboards capture what the system has been designed to count.

Users experience the entire relationship.

They experience delays between steps.

Repeated requests for information.

Unclear consequences.

Forced account creation.

Notification pressure.

Loss of control.

Accumulated friction across multiple products and services.

Each event may appear insignificant in isolation.

The total experience may feel exhausting.

A dashboard can show successful interactions while missing the burden created by their accumulation.

The Measurement Boundary

Every product measurement system has a boundary.

It records some effects and excludes others.

A platform may measure clicks but not regret.

A service may measure task completion but not cognitive effort.

A marketplace may measure purchases but not declining trust.

An application may measure daily use but not the user's desire to escape it.

What falls outside the measurement boundary becomes harder to defend in product decisions.

The absence of a negative metric can be mistaken for the absence of harm.

Goodhart's Law and Product Design

Goodhart's Law is often summarized as the idea that when a measure becomes a target, it stops being a good measure.

In product development, the mechanism is often gradual.

The metric remains useful.

It still responds to real behavior.

But the organization becomes increasingly skilled at producing the behavior the metric records.

The relationship between the behavior and genuine value weakens.

The metric has not become meaningless.

It has become easier to produce without producing the full outcome it once represented.

When Research Becomes a Validation Layer

User research can correct metric-driven drift.

It can also become subordinate to it.

Research questions may be framed around explaining dashboard movement.

Interviews may focus on validating a proposed feature.

Usability studies may evaluate whether users can complete a predefined flow.

The user remains present.

But the range of possible feedback has already been narrowed by the representation the organization is trying to improve.

Research becomes a way of refining the proxy rather than reconsidering it.

Representational Optimization

Representational Optimization occurs when the organization improves what can be seen through its measurement system.

The dashboard becomes cleaner.

The funnel becomes more efficient.

The user journey becomes more standardized.

The performance report becomes more convincing.

The product appears increasingly legible from inside the organization.

That internal legibility can grow while the external experience becomes less coherent.

The organization learns more about the representation.

It does not necessarily learn more about the user.

Why the System Still Looks Healthy

Experience Drift is difficult to detect because the standard indicators continue signaling success.

Conversion rises.

Retention remains stable.

Onboarding completion improves.

Support tickets decline.

Feature adoption increases.

Experiment velocity accelerates.

These are real operational achievements.

They do not prove that the product remains aligned with the people using it.

A system can become increasingly effective at producing measurable behavior while becoming less useful, humane, or trustworthy.

The absence of metric failure is not proof of experiential alignment.

Semantic Fidelity in Product Metrics

Semantic Fidelity describes how well a representation preserves the meaning of what it was created to represent.

A metric has high semantic fidelity when movement in the number still corresponds to meaningful improvement in the experience.

It has low semantic fidelity when the number can improve while the underlying outcome remains unchanged or worsens.

The central product question is therefore not only:

Did the metric move?

It is also:

Does this metric still mean what we think it means?

Metrics require continual contact with the experience that originally gave them significance.

A Recognition Test

Ask:

Can this metric improve without the user receiving more value?

What important part of the experience falls outside the dashboard?

Are users behaving this way because the product helps them or because the interface pressures them?

Do qualitative reports conflict with quantitative performance?

Has the organization become better at moving the metric than explaining why the metric matters?

Would the product still be considered successful if the dashboard were unavailable?

These questions test whether measurement still supports understanding or has begun replacing it.

Product Metrics and Reality Drift

Within the **Reality Drift framework**, product management operates through representations of experience.

Metrics, surveys, personas, journey maps, experiments, and dashboards allow organizations to make user experience visible at scale.

As those representations become more influential, teams increasingly optimize the product through them.

Reality Drift begins when the representations remain coherent and successful while becoming less answerable to the experience they were created to reveal.

The dashboard improves.

The user adapts, struggles, or leaves elsewhere.

The Core Principle

Product metrics are valuable because they make experience visible.

They become dangerous when visibility is mistaken for completeness.

Product metrics can improve while the product feels worse because the organization is optimizing a measurable representation of experience rather than the experience as a whole.

Experience Drift begins when dashboard success remains visible but the user's reality becomes increasingly difficult to recover from the data.

Proxy Optimization does not require false metrics. It only requires a system that becomes better at producing the signal than preserving the outcome behind it.

The goal is not to stop measuring users. It is to keep every measurement answerable to the experience it claims to represent.

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